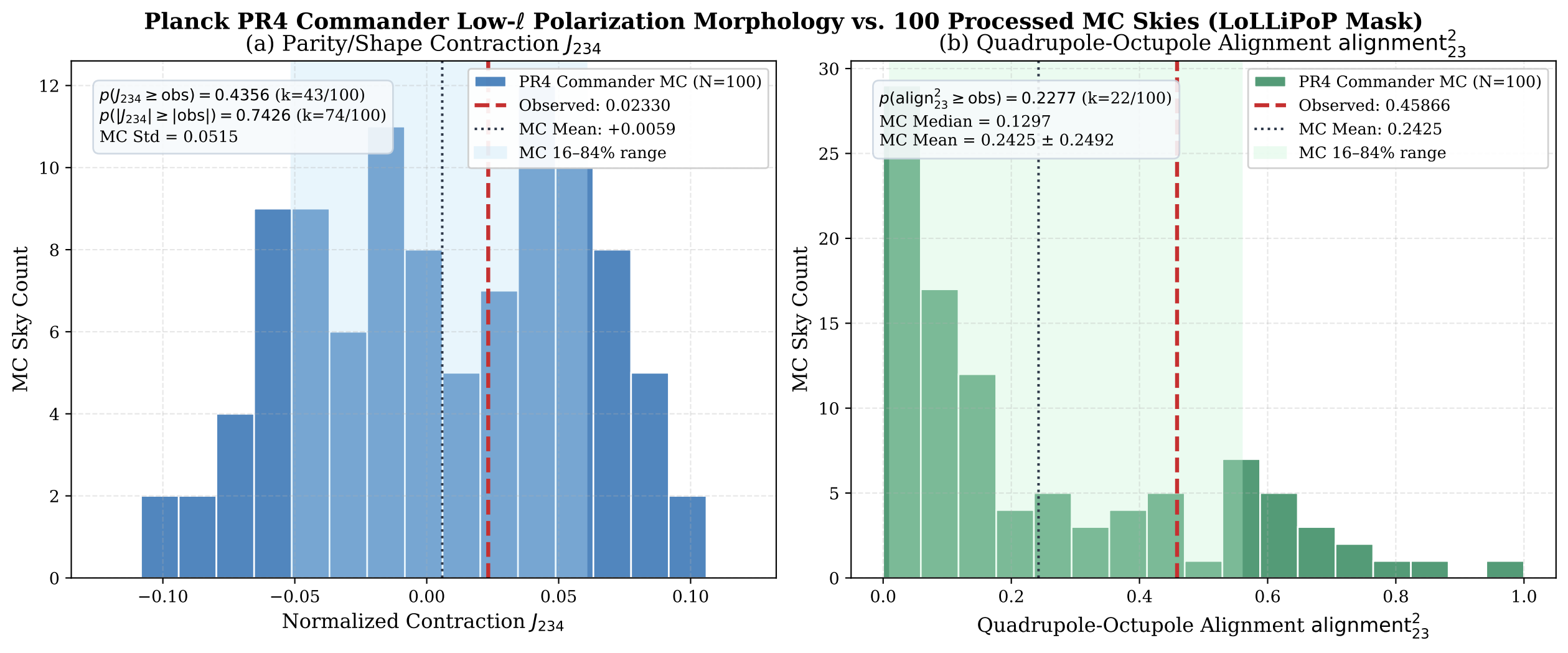
Planck PR4 Commander Low-ℓ Polarization Morphology vs. 100 Processed MC Skies (LoLLiPoP Mask)
(a) Parity/Shape Contraction J_{234}
$p(J_{234} \geq \text{obs}) = 0.4356$ (k=43/100)
$p(|J_{234}| \geq |\text{obs}|) = 0.7426$ (k=74/100)
MC Std = 0.0515
PR4 Commander MC (N=100)
Observed: 0.02330
MC Mean: +0.0059
MC 16–84% range
MC Sky Count
Normalized Contraction J_{234}
(b) Quadrupole-Octupole Alignment alignment^2_{23}
$p(\text{align}^2_{23} \geq \text{obs}) = 0.2277$ (k=22/100)
MC Median = 0.1297
MC Mean = 0.2425 ± 0.2492
PR4 Commander MC (N=100)
Observed: 0.45866
MC Mean: 0.2425
MC 16–84% range
MC Sky Count
Quadrupole-Octupole Alignment alignment^2_{23}